K-means Clustering Analysis of Student Grades in Math, English, and Science

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Loading and Preparing Data

```
grades <- read.csv("/Users/admin/Downloads/grades.csv")
grades_data <- grades[,-1]
```

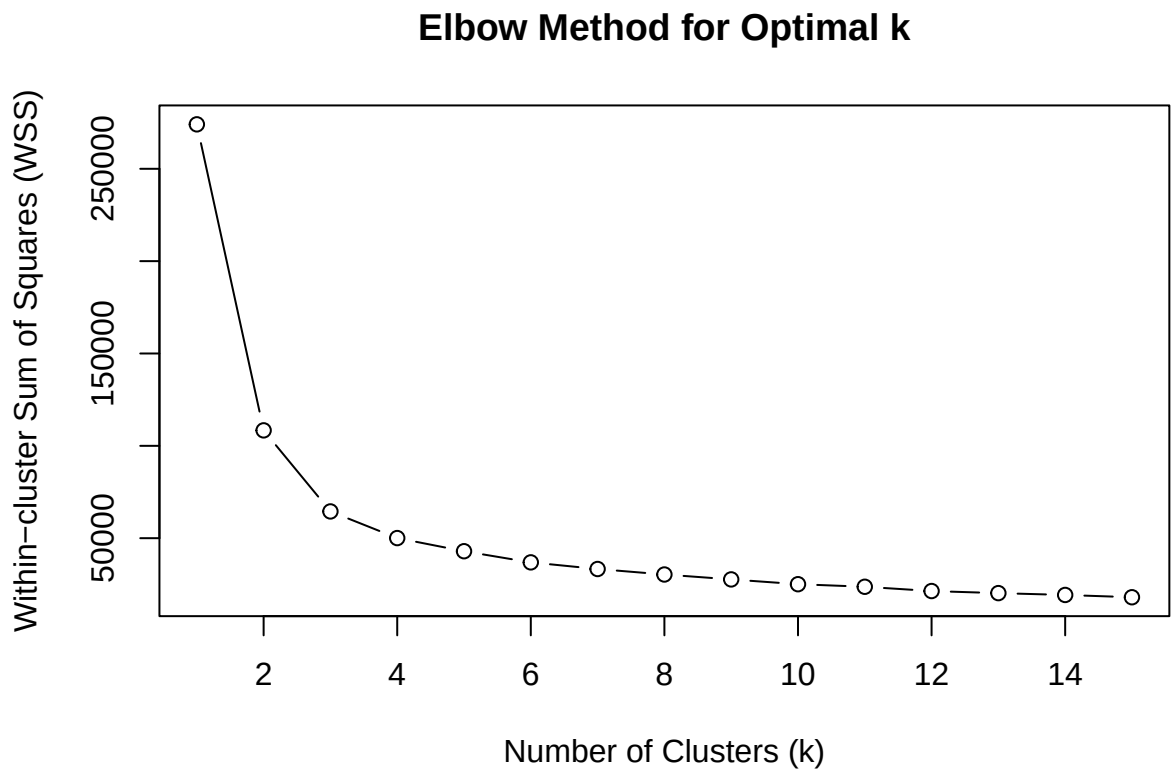
Calculate WSS for k = 1 to 15

```
calculate_wss <- function(data, k) {
  km <- kmeans(data, centers = k, nstart = 25)
  return(km$tot.withinss)
}

wss <- sapply(1:15, function(k) calculate_wss(grades_data, k))
```

Elbow Curve for Optimal k

```
plot(1:15, wss, type = "b",
     xlab = "Number of Clusters (k)",
     ylab = "Within-cluster Sum of Squares (WSS)",
     main = "Elbow Method for Optimal k")
```



This elbow plot shows that the curve starts with a steep decline and then levels off. The WSS value begins at approximately 25,000 for k=1 and drops dramatically until around k=3.The elbow occurs around k=3, thus indicating the optimal number of clusters for this dataset.

K-means Clustering

```
optimal_k <- 3
km_result <- kmeans(grades_data, centers = optimal_k, nstart = 25)
grades$Cluster <- as.factor(km_result$cluster)
```

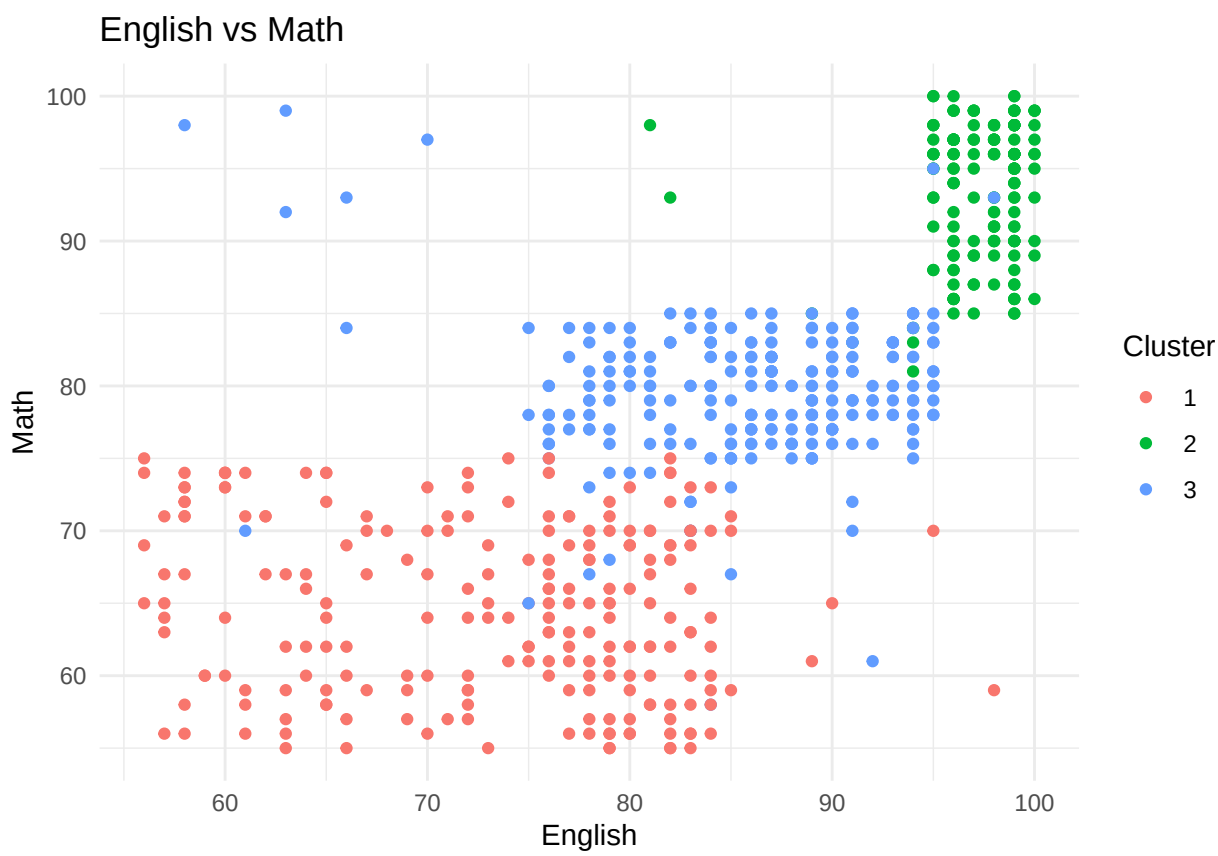
Scatter Plot Function

```
plot_scatter <- function(data, x, y) {
  ggplot(data, aes(x = .data[[x]], y = .data[[y]], color = Cluster)) +
    geom_point() +
    labs(x = x, y = y, title = paste(x, "vs", y)) +
    theme_minimal()
}
```

Generating Plots

```
p1 <- plot_scatter(grades, "English", "Math")
p2 <- plot_scatter(grades, "English", "Science")
p3 <- plot_scatter(grades, "Math", "Science")
```

English vs Math

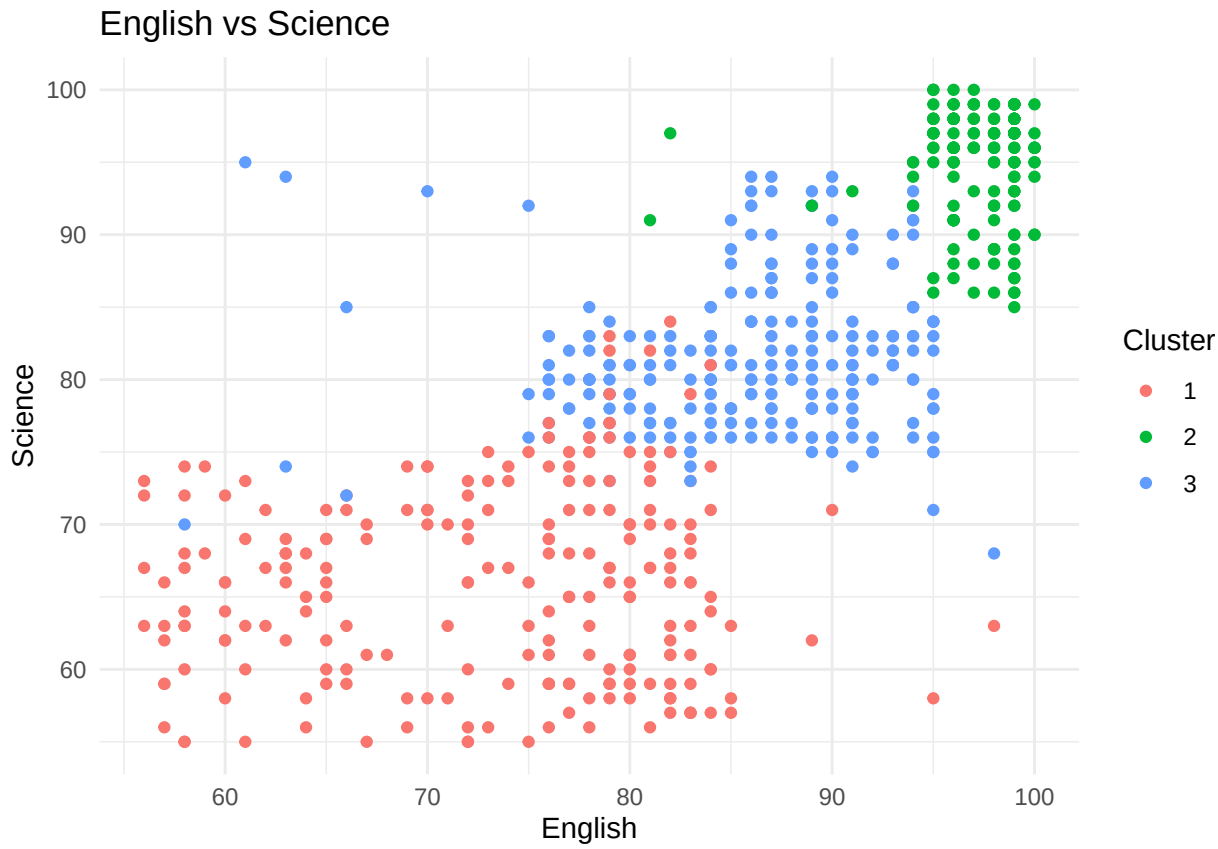


Red Cluster: This group shows scores ranging from English: 60-80 points for English, and 55-75 points for Math.This cluster represents students who scored the lowest in both subjects.

Green Cluster: This cluster shows students with scores ranging from 90-100 points for both English and Math. This cluster represents the highest scoring students, showing strong performance in both subjects.

Blue Cluster: This cluster shows students with scores ranging from 75-95 points for English and 75-90 points for Math. It represents the average to above-average performing students.

English vs Science

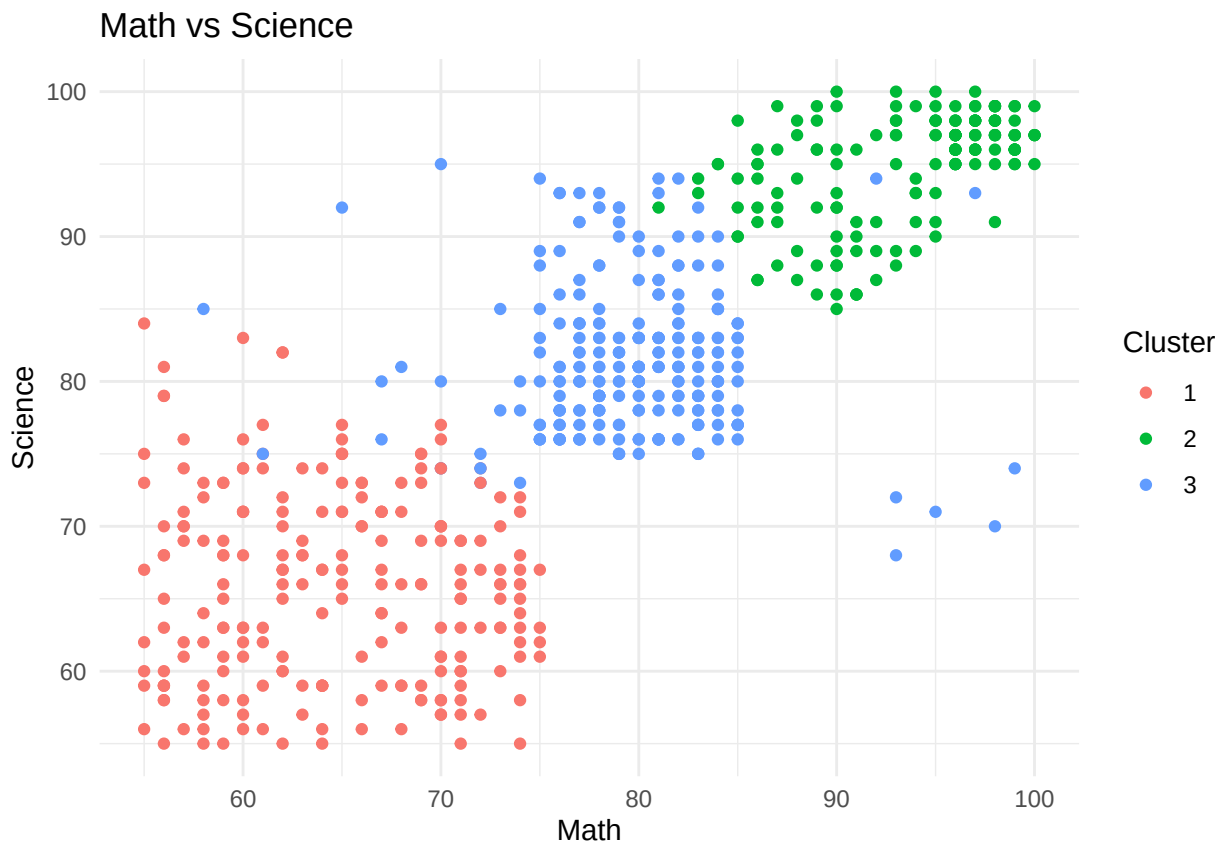


Red Cluster: Shows the widest spread of scores with students scoring between 60-70 in English and 60-75 in Science. This cluster shows generally lower performance in both subjects.

Blue Cluster: This cluster shows moderate variability with scores being around 75-85 in both English and Science. The cluster demonstrates average to above average performance in both subjects.

Green Cluster: This cluster shows scores of around 90-100 in both subjects and represents exceptional performance in both English and Science.

Math vs Science



Red Cluster: This shows Math scores ranging approximately from **55-70**, and Science scores ranging approximately from **55-80**. The cluster represents students with lower performance in both Math and Science.

Green Cluster: This cluster shows Math scores ranging approximately from 85-100, and Science scores ranging approximately from 85-100. The cluster represents high-performing students excelling in both Math and Science.

Blue Cluster: This cluster shows Math scores ranging approximately from 75-85 and Science scores ranging approximately from 75-90.It represents students with average performance in both subjects